

Theorem 4.3 [15] *If G is a unicyclic graph of order $n \geq 3$ and girth $g \geq 3$, then*

$$rx_k(G) = \begin{cases} n-2 & \text{if } k=3 \text{ and } g \geq 4, \\ n-1 & \text{if } g=3 \text{ or } 4 \leq k \leq n. \end{cases}$$

The investigation of $rx_k(G)$ for a general k and a general graph G is rather difficult. So one may consider $rx_k(G)$ for a special graph class, or, for a general graph and small k , such as $k=3$.

Problem 4.4 *Derive a sharp upper bound for $rx_3(G)$.*

There is also a generalization of the k -rainbow index, say (k, ℓ) -rainbow index $rx_{k,\ell}$, of G which is mentioned in [15] and we will not introduce it here.

5 Rainbow vertex-connection number

The above several parameters are all defined on edge-colored graphs. Here we will introduce a new graph parameter which is defined on vertex-colored graphs. It is, as mentioned above, a vertex-version of the rainbow connection number. Krivelevich and Yuster [36] put forward this new concept and proved a theorem analogous to Theorem 2.17.

Theorem 5.1 [36] *A connected graph G with n vertices has $rvc(G) < \frac{11n}{\delta(G)}$.*

The argument of this theorem used the concept of k -strong two-step dominating sets. They proved that if H is a connected graph with n vertices and minimum degree δ , then it contains a $\frac{\delta}{2}$ -strong two-step dominating set S whose size is at most $\frac{2n}{\delta+2}$. Then they derived an edge-coloring for G according to its connected $\frac{\delta}{2}$ -strong two-step dominating set. And they showed that, with positive probability, their coloring yields a rainbow connected graph by the Lovász Local Lemma (see [3]).

Motivated by the method of Theorem 5.1, Li and Shi derived an improved result.

Theorem 5.2 [40] *A connected graph G of order n with minimum degree δ has $rvc(G) \leq 3n/(\delta+1) + 5$ for $\delta \geq \sqrt{n-1} - 1, n \geq 290$, while $rvc(G) \leq 4n/(\delta+1) + 5$ for $16 \leq \delta \leq \sqrt{n-1} - 2$, $rvc(G) \leq 4n/(\delta+1) + C(\delta)$ for $6 \leq \delta \leq 16$ where $C(\delta) = e^{\frac{3\log(\delta^3+2\delta^2+3)-3(\log 3-1)}{\delta-3}} - 2$, $rvc(G) \leq n/2 - 2$ for $\delta = 5$, $rvc(G) \leq 3n/5 - 8/5$ for $\delta = 4$, $rvc(G) \leq 3n/4 - 2$ for $\delta = 3$. Moreover, an example shows that when $\delta \geq \sqrt{n-1} - 1$, and $\delta = 3, 4, 5$ the bounds are seen to be tight up to additive factors.*

Motivated by the method of Theorem 5.1, Dong and Li [22] also proved a theorem analogous to Theorem 2.21 for the rainbow vertex-connection version according to the degree sum condition σ_2 , which is stated as the following theorem.

Theorem 5.3 [22] *For a connected graph G of order n , $rc(G) \leq 8\frac{n-2}{\sigma_2+2} + 10$ for $2 \leq \sigma_2 \leq 6$ and $\sigma_2 \geq 28$, while for $7 \leq \sigma_2 \leq 8$ and $16 \leq \sigma_2 \leq 27$, $rc(G) \leq \frac{10n-16}{\sigma_2+2} + 10$, and for $9 \leq \sigma_2 \leq 15$, $rc(G) \leq \frac{10n-16}{\sigma_2+2} + A(\sigma_2)$, where $A(\sigma_2) = 63, 41, 27, 20, 16, 13, 11$, respectively.*

Note that by the definition of σ_2 , we know $\sigma_2 \geq 2\delta$, so we have $8\frac{n-2}{\sigma_2+2} + 10 \leq \frac{4(n-2)}{\delta+1} + 10$, and hence the bound of Theorem 5.3 is an improvement to that of Theorem 5.2 in the case $16 \leq \delta \leq \sqrt{n-1} - 2$.

6 Algorithms and computational complexity

At the end of [9], Caro, Lev, Roditty, Tuza and Yuster gave two conjectures (see Conjecture 4.1 and Conjecture 4.2 in [9]) on the complexity of determining the rainbow connection numbers of graphs. Chakraborty, Fischer, Matsliah and Yuster [10] solved these two conjectures by the following theorem.

Theorem 6.1 [10] *Given a graph G , deciding if $rc(G) = 2$ is NP-Complete. In particular, computing $rc(G)$ is NP-Hard.*

Chakraborty, Fischer, Matsliah and Yuster divided the proof of Theorem 6.1 into three steps: the first step is showing the computational equivalence of the problem of *rainbow connection number 2*, that asks for a red-blue edge coloring in which all vertex pairs have a rainbow path connecting them, to the problem of *subset rainbow connection number 2*, that asks for a red-blue coloring in which every pair of vertices in a given subset of pairs has a rainbow path connecting them. In the second step, they reduced the problem of *extending to rainbow connection number 2*, asking whether a given partial red-blue coloring can be completed to obtain a rainbow connected graph, to the problem of subset rainbow connection number 2. Finally, the proof of Theorem 6.1 is completed by reducing 3-SAT to the problem of extending to rainbow connection number 2.

Chakraborty, Fischer, Matsliah and Yuster [10] also raised the following problem.

Problem 6.2 [10] *Suppose that we are given a graph G for which we are told that $rc(G) = 2$. Can we rainbow-color it in polynomial time with $o(n)$ colors?*

For the usual coloring problem, this version has been well studied. It is known that if a graph is 3-colorable (in the usual sense), then there is a polynomial time algorithm that colors it with $\tilde{O}(n^{3/14})$ colors [5].

Suppose we are given an edge coloring of the graph. Is it then easier to verify whether the colored graph is rainbow connected? Clearly, if the number of colors is constant then this problem becomes easy. However, if the coloring is arbitrary (with an unbounded number of colors), the problem becomes NP-Complete.

Theorem 6.3 [10] *The following problem is NP-Complete: Given an edge-colored graph G , check whether the given coloring makes G rainbow connected.*

For the proof of Theorem 6.3, Chakraborty, Fischer, Matsliah and Yuster first showed that the $s - t$ version of the problem is NP-Complete. That is, given two vertices s and t of an edge-colored graph, decide whether there is a rainbow path connecting them. Then they reduced the problem of Theorem 6.3 from it.

More generally it has been shown in [37], that for any fixed $k \geq 2$, deciding if $rc(G) = k$ is NP-complete.

In [10], Chakraborty, Fischer, Matsliah and Yuster also derived some positive algorithmic results. Parts of the following two results were shown in Theorems 2.16 and 2.48. They proved:

Theorem 6.4 [10] *For every $\epsilon > 0$ there is a constant $C = C(\epsilon)$ such that if G is a connected graph with n vertices and minimum degree at least ϵn , then $rc(G) \leq C$. Furthermore, there is a polynomial time algorithm that constructs a corresponding coloring for a fixed ϵ .*

As mentioned above, Theorem 6.4 is based upon a modified degree-form version of Szemerédi's Regularity Lemma that they proved and that may be useful in other applications. From their algorithm it is also not hard to find a probabilistic polynomial time algorithm for finding this coloring with high probability (using on the way the algorithmic version of the Regularity Lemma from [2] or [26]).

Theorem 6.5 [10] *If G is an n -vertex graph with diameter 2 and minimum degree at least $8 \log n$, then $rc(G) \leq 3$. Furthermore, such a coloring is given with high probability by a uniformly random 3-edge-coloring of the graph G , and can also be found by a polynomial time deterministic algorithm.*

As mentioned, He and Liang [29] investigated the rainbow k -connectivity in the setting of random graphs. They also investigated rainbow k -connectivity from the algorithmic point of view. The NP-hardness of determining $rc(G)$ was shown by Chakraborty et al. as shown above. They showed that the problem (even the search version) becomes easy in random graphs.

Theorem 6.6 [29] *For any constant $\epsilon \in [0, 1)$, $p = n^{-\epsilon(1 \pm o(1))}$ and $k \leq O(\log n)$, there is a randomized polynomial time algorithm that, with probability $1 - o(1)$, makes $G(n, p)$ rainbow- k -connected using at most one more than the optimal number of colors.*

Since almost all natural edge probability functions p encountered in various scenarios have such form, their result is quite strong. Note that $G(n, n^{-\epsilon})$ is almost surely disconnected